

Manuel Mezo

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Links: [Linkedin](#) | [Github](#) | [Web CV and project portfolio](#)

Profile

Computer Vision and Robotics Builder and engineer blending deep technical expertise in custom **ML models**, **ROS2**, and **computer vision** with an 8+ year track record of solving **multi-million dollar operational challenges** at scale for Amazon and McKinsey. Now focused on independently developing and deploying functional **robotic systems** and **AI vision applications**, bringing a combination of **hardware/software engineering capability** and high-level operational rigor to the **AI and Robotics industry**.

Technical Skills

Robotics & Simulation: ROS2, Gazebo, CoppeliaSim, Arduino, Inverse Kinematics, 3D Printing, CAD Design; **Computer Vision & ML:** OpenCV, YOLO (open-source fine-tuning), Ultralytics, Roboflow, ByteTrack, Vision tracking/counting/zone-based logic; **Development & Execution:** Python (Advanced), C++, SQL (Advanced), Git/GitHub, CI/CD pipelines; **AI, Cloud & Data:** Google Cloud, AWS, Pandas, NumPy, Tableau, Power BI

Computer Vision & Robotics Projects

- **Custom YOLOv8 Object Tracking & Spatial Logic:** Engineered a **custom object detection pipeline** using an annotated dataset in **Roboflow** with augmentation for motion blur and lighting. **Fine-tuned a YOLOv8n model (~97% mAP@50)** and implemented **real-time multi-object tracking** via **ByteTrack** and **OpenCV**. Developed **advanced spatial logic** for moving conveyor belts, featuring **point-in-polygon virtual pick-zones** and **vertical throughput tripwires**.
- **Integrated VLA AI model in 3D Printed Robotic Arm:** Built a **physical 4-axis robotic arm** integrated with a **cloud-based Vision-Language-Action (VLA) reasoning engine**. The system utilizes **Gemini 1.5 Pro** to translate **natural language voice commands** into precise **local mechanical actuations**.
- **Self-Balancing Robot & ROS2 Navigation:** Developed a **physical self-balancing robot** deployed on **Arduino hardware**, successfully implementing **IMU filtering**, **motor actuation**, and **PID control loops**. Additionally, built a **ROS2 simulation** in **Gazebo** focusing on **sensor modeling**, **odometry**, and **multithreaded Python scripts** to parse **LIDAR data** for **autonomous wall-following navigation**.
- **VisionAI & Real-Time Traffic Analyzer:** Developed a browser-based dashboard for **real-time video telemetry**, allowing users to define virtual gates to classify and count vehicles/pedestrians. Built **VisionAI**, an application powered by **Gemini 2.5 Flash** that **detects and tracks objects** via **natural language queries** with **live bounding boxes**.

Professional Experience

Manager, EU Transportation Forecasting | Amazon (Madrid, Spain)

Feb 2026 – Present

- Own the weekly production of end-to-end aligned demand forecasts to **optimize transportation** and labor capacity planning across all EU network segments.
- Drive **continuous improvement** of the **forecast generation process**, overseeing **simulation modules**, **ML bias correction**, and automated health checks.

- Co-own a strategic roadmap with the **Forecasting Tech team** to deliver features aiming for a 150 bps inaccuracy (WAPE) reduction, representing **>\$10M/year in cost-savings**.
- Manage a distributed team of 6 Business Intelligence Engineers (BIEs) and Program Managers.

Sr Program Manager, Supply Chain Planning | Amazon (Madrid, Spain)

Sept 2023 – Feb 2026

- Invented and implemented a long-range (52-week) forecasting methodology for EU truckload demand, supporting S&OP cycles and driving **\$140Mn CapEx decisions** for 2025.
- Extended the forecasting horizon for inbound flows to 12 weeks, improving planning for **800k annual labor hours**.
- Improved **inventory transfer truck fill rate modeling** (inaccuracy reduced from 35% to 17%), **saving ~\$1.8Mn annually** by preventing late additions and cancellations.
- Identified and resolved a **critical logic defect** in the automated scheduling system using SQL-based data analysis and Python scripting, protecting **millions in extra operational costs**.

Operations Consultant | McKinsey & Company (Madrid, Spain)

Aug 2022 – Jul 2023

- Redesigned and standardized the production capacity planning tool and process of a global pharmaceutical manufacturer through an automated, KPI-oriented model and rolled it out in 6 plants (13 production lines).
- Optimized production batch sizes across 9 plants, saving \$800k/year (1.1% of conversion cost) of a global pharmaceutical manufacturer.
- Optimized the production plan and timelines of a trains manufacturer recovering 2.5 months of delivery delays worth \$1Mn in saved penalties.

Sr Program Manager, 3PL Ops Excellence & Quality | Amazon (Madrid, Spain & Luxembourg)

Sept 2017 – Aug 2022

- Led the planning and execution of multi-item order shipping capabilities across 10 3PL sites, generating **\$10Mn+ in transport savings**.
- Designed network-wide **inventory audit programs** across 25+ warehouses, improving the **Inventory Record Defect Rate metric by 50%**.

Education & Certifications

- **MSc in Aerospace Engineering** | Universidad Carlos III de Madrid (*Included Master's Thesis Exchange at TU Delft in The Netherlands*)
- **BSc in Aerospace Engineering** | Universidad de León (2011 – 2015)
- **BSc in Mechanical Engineering** | Universidad de León (2013 – 2016)
- **Certificate of Proficiency: ROS2 Basics (Python)** | The Construct Robotics Institute (2026)